

TD02 – Python + CSV / JSON

Part 1 - Review the starter

```
((base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
```

1. Which function reads JSON?

```
def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)
```

2. Which function reads CSV?

```
def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))
```

3. Which function selects one ticker?

```
def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]
```

4. Which function coordinates execution?

```
def main():
```

```
    instruments = load_instruments()
    prices = load_prices()
```

5. Where are the input files stored?

Part 2 - Understand JSON and dictionaries

```
print(type(instruments))
print(instruments.keys())
```

```
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
```

```
instrument = instruments["instrument"]
benchmark = instruments["benchmark"]

print(instrument["ticker"])
print(instrument["name"])
print(instrument["currency"])

print(benchmark["ticker"])
print(benchmark["name"])
```

```
AAPL
Apple Inc.
USD
SP500
S&P 500
```

Part 3 - Understand CSV and lists of dictionaries

```
prices = load_prices()

print(type(prices))
print(type(prices[0]))
print(prices[0])
```

```
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
```

```
print(prices[0]["close"])
print(type(prices[0]["close"]))

close = float(prices[0]["close"])
print(close)
print(type(close))
```

```
250.00
<class 'str'>
250.0
<class 'float'>
```

Part 4 - Filter instrument and benchmark rows

```
print(len(instrument_prices))
print(len(benchmark_prices))
```

```
21
21
```

Part 5 - Create first and last close helpers

```
def get_first_close(prices):
    return float(prices[0]["close"])
```

```
print(f"{instrument['ticker']} -> {get_first_close(instrument_prices)}")
print(f"{benchmark['ticker']} -> {get_first_close(benchmark_prices)}")
```

```
AAPL -> 250.0
SP500 -> 6600.0
```

```
def get_last_close(prices):
    return float(prices[-1]["close"])
```

```
print(f"{instrument['ticker']} -> {get_last_close(instrument_prices)}")
print(f"{benchmark['ticker']} -> {get_last_close(benchmark_prices)}")
```

```
AAPL -> 266.2
SP500 -> 6742.0
```

Part 6 - Build one reusable market summary

```
def display_market_summary(asset, prices, show_currency=True):
    latest_price = prices[-1]
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

    print(f"{asset['ticker']} - {asset['name']}")
    if show_currency:
        print(f"Last price: {latest_price['close']} {asset['currency']}")
    else:
        print(f"Last level: {latest_price['close']}")

    print(f"Observations: {len(prices)}")
    print(f"First close: {first_close}")
    print(f"Last close: {last_close}")
    print()
```

```
display_market_summary(instrument, instrument_prices)
display_market_summary(benchmark, benchmark_prices)
```

```
AAPL - Apple Inc.  
Last price: 266.20 USD  
Observations: 21  
First close: 250.0  
Last close: 266.2  
  
SP500 - S&P 500  
Last price: 6742.00 USD  
Observations: 21  
First close: 6600.0  
Last close: 6742.0
```

Part 7 - Keep main() readable

```
def main():  
    instruments = load_instruments()  
    prices = load_prices()  
  
    instrument = instruments["instrument"]  
    benchmark = instruments["benchmark"]  
  
    instrument_prices = filter_prices(prices, instrument["ticker"])  
    benchmark_prices = filter_prices(prices, benchmark["ticker"])  
  
    print("=== MarketPulse ===")  
    print()  
    print("Market configuration")  
    print(f"Period : {LOOKBACK_LABEL}")  
    print(f"Interval : {INTERVAL_LABEL}")  
    print()  
    print("Instrument")  
    display_market_summary(instrument, instrument_prices)  
    print()  
    print("Benchmark")  
    display_market_summary(benchmark, benchmark_prices)
```

Part 8 - Validate the final output

```
((base) daphnecortesse@MacBook-Air-de-Daphne esilv-marketpulse-g07-nantais % python src/main.py  
=== MarketPulse ===  
  
Market configuration  
Period : 1 month  
Interval : Daily  
  
Instrument  
AAPL - Apple Inc.  
Last price: 266.20 USD  
Observations: 21  
First close: 250.0  
Last close: 266.2  
  
Benchmark  
SP500 - S&P 500  
Last price: 6742.00 USD  
Observations: 21  
First close: 6600.0  
Last close: 6742.0
```